1. **Reset your root password on VM1 and make it 'idtech24'**
   Rb.break and init=/bin/bash

2. **Configure your hostname, ip address, gateway and dns.**
   a. Hostname: idtech
   b. IP adress: 192.168.x.170/24
   c. Gateway: Look your current gateway
   d. DNS: 8.8.8.8

3. **Configure YUM repos.** Do it by mounting your sr0 device to mnt folder. Before configuring delete all default repos from device. Do it by one command (I will look your history) (2 repos: 1st is Base and 2nd is AppStream). Install Apache.

4. Configure your system to use Flatpak repositories.

   On your system, configure Flatpak so that user **develop** can install the Calculator from the Flatdb repository.

The following requirements must be met:

- A Flatpak repository has been made available from https://dl.flathub.org/repo/flathub.flatpakrepo
- The Flatpak repository should be named **flatdb**.
- The repository must be configured only for the user **develop**.
- Install the **Calculator** application for user **develop**.

5. **Install apache web server. Change default port to 9090 and open in browser**. A web server running on non standard port 9090 is having issues serving content.
   Debug and fix the issues.
   Web service should automatically start at boot time.

6. **Set default values for new users.** Set the default password validity to 60 days, and set the first UID that is used for new users to 3500.

7. Create a collaborative DIR. (change group owner, set the permissions along with sgid)

   - Create the Directory "/home/linux" with the following characteristics. - Group ownership of "/home/linux" should go to "data" group.
   - The directory should have full permission for all members of "data" group but not to the other users except "root".
   - Files created in future under "/home/linux" should get the same group ownership.

8. **Create User accounts with supplementary (secondary) group.**
   - Create the group a named "data".
   - Create users as named "gandi" and "issak", will be the supplementary group "data".
   - Create a user as named "magellan", should have non-interactive shell and it should be not the member of "data".
   - Password for all users should be "Tro123tent"

9. **Create a logical Volume and mount it permanently:**
   - Create 2Gb volume group named idtechvolume. Create the logical volume with the name "userlvm" by using 35%FREE from the volume group "idtechvolume".
   - Consider each PE size of the volume group as "32 MB".
   - Create /LVM folder and mount it there with file system xfs.

10. **Resize a logical Volume:**

    Resize the logical volume to 700MiB "userlvm" so that after reboot the size should be in between 800MB to 950MB.

11. Create LVM and add a swap partition of 750MB and mount it permanently.

12. **Copy the file /etc/fstab to /var/tmp/ and configure the "ACL" as mentioned following.**

- The file /var/tmp/fstab should be owned by the "root".
- The file /var/tmp/fstab should belong to the group "root".
- The file /var/tmp/fstab should not be executable by any one. ● The user "magellan" should be able to read and write to the file.
- The user "gandi" can neither read nor write to the file.

13. Your system to synchronize the time from “az.pool.ntp.org”

14. Create user 'bob' with 3576 uid and set the password 'troo123tent'.

15. Create an archive '/root/backup.tar.bz2' of /usr/local directory and compress it with bzip2.

On VM2

16. Configure Recurring jobs

a) Create a script /usr/local/bin/rhcsa that lists all files in /home and saves the output to /root/log_output/home_logs.trc. Make sure the script is executable
b) Create a systemd service unit **rhcsa.service** that runs the script
c) Create a systemd timer unit **rhcsa.timer** that triggers the service every 1 minutes.
d) Enable and start the timer so it begins working immediately and continues after reboots.
e) Verify the timer is active and confirm that /root/log_output/home_logs.trc is updated

17. **Configure a cron job:**

a) that runs every weekend at 18:01 and executes: logger "EX200 in progress" as the user gandi;

b) Configure a cron job for user "issak", cron must runs daily at 3:42am and inside executes the /usr/bin/echo "Hello 2026"

18. Find the rows that contain only “Cmnd_Alias” from /etc/ and write result to the /tmp/sudo.txt

19. Configure NFS server, export /nfsshared folder in your VM2:
    a. This folder must be writeable.
    b. Share this folder only for your client computer.
    c. On client computer that folder should be automount autofs.
    d. Go into “files” folder in the “nfsshared” folder using autofs.

20. Customize tuned profile and set powersave.